

Determine the asymptotically tight rainbow cycle number in bounded-partite digraphs

ChatGPT (gpt-5.6-sol)*

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Abstract

For a positive integer d , the rainbow cycle number $R(d)$ is defined in terms of directed multipartite graphs whose nonempty vertex classes have size at most d and in which every vertex has an in-neighbor in every other class. We prove the packing inequality

$$(k-1)! \sum_{i=1}^k |V_i| \leq \prod_{i=1}^k |V_i|$$

for every such k -partite graph that contains no rainbow directed cycle. It follows that $k! \leq d^{k-1}$, and consequently that $R(d) < ed$. Combined with the standard construction showing $R(d) \geq d$, this yields

$$d \leq R(d) < ed,$$

and hence $R(d) = \Theta(d)$.

1 The open problem

This paper addresses an open problem posed by Bhaskar Ray Chaudhury, Jugal Garg, Kurt Mehlhorn, Ruta Mehta, Pranabendu Misra in *Improving Envy Freeness up to Any Good Guarantees Through Rainbow Cycle Number* [1] (Mathematics of Operations Research, 2023), Page 5, end of Section 1.2 (immediately after the discussion of $\{(R(2) \leq 2)\}$); reiterated on page 16 right after Theorem 6 in Section 5. The website catalogue entry is problem 98 (W4388912937_p0) of the [Open Problems in Operations Research](#) collection.

Determine asymptotically tight upper bounds on $R(d)$ as a function of d . In particular, decide whether $R(d) = O(d)$ holds (equivalently, whether there exists a constant C such that $R(d) \leq Cd$ for all $d \geq 1$).

*Generated by the `base_solver_non_interactive` (MOR sweep) pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.9 write-up; run `run_20260826_172911_9e388b55`, `sweep-audit-approved`). Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

2 Introduction

The rainbow cycle number was introduced by Chaudhury, Garg, Mehlhorn, Mehta, and Misra in connection with partial approximate envy-freeness up to any good (EFX) allocations [2]. They established

$$d \leq R(d) \leq d^4 + d$$

and conjectured that $R(d)$ grows linearly in d .

The main ingredient of this note is a packing inequality, obtained by associating a transversal with every ordering of the vertex classes and every choice of a terminal vertex. If two distinct orderings produced the same transversal, the corresponding directed paths would create a rainbow cycle. The resulting injection yields an exact counting inequality and, in particular, the upper bound $R(d) < ed$.

Regarding attribution, the packing inequality and essentially the same injection proof appear as Theorem 3 of Sivashankar [3]. The present note is therefore intended as a self-contained exposition rather than as a claim of new results.

3 Definitions

A *directed k -partite graph* is a directed graph G whose vertex set is partitioned into nonempty classes

$$V(G) = V_1 \sqcup \cdots \sqcup V_k,$$

with every edge joining vertices in different classes.

A directed cycle is understood to be simple and to have length at least two; directed 2-cycles are allowed.

Definition 1. A directed cycle in G is *rainbow* if it contains at most one vertex from each class V_i .

Thus a rainbow cycle need not meet every class.

Definition 2. Let d be a positive integer. The *rainbow cycle number* $R(d)$ is the largest integer k for which there exists a directed k -partite graph G with nonempty classes $V_1, \dots, V_k$ satisfying the following conditions:

1. $|V_i| \leq d$ for every i ;
2. for every $i \neq j$ and every $v \in V_j$, there is some $u \in V_i$ such that $u \rightarrow v$ is an edge;
3. G contains no rainbow directed cycle.

The second condition will be called the *cross-part in-neighbor condition*.

4 The packing inequality

We first prove a bound that retains the individual class sizes.

Theorem 1 (Packing inequality). *Let G be a directed k -partite graph with nonempty classes $V_1, \dots, V_k$, and write $n_i = |V_i|$. If G satisfies the cross-part in-neighbor condition and contains no rainbow directed cycle, then*

$$(k-1)! \sum_{i=1}^k n_i \leq \prod_{i=1}^k n_i. \quad (1)$$

Proof. For every $i \neq j$ and every $v \in V_j$, fix a vertex $\phi_{i,j}(v) \in V_i$ such that $\phi_{i,j}(v) \rightarrow v$ is an edge. Only these selected edges will be used below; equivalently, we may replace G by the spanning subgraph consisting of the selected edges. Removing edges preserves the cross-part in-neighbor condition and cannot create a rainbow cycle.

Let $\pi = (\pi_1, \dots, \pi_k) \in \mathfrak{S}_k$ be a permutation of the classes, and choose a terminal vertex $z \in V_{\pi_k}$. Define vertices $v_k, \dots, v_1$ recursively by

$$v_k = z, \quad v_t = \phi_{\pi_t, \pi_{t+1}}(v_{t+1}) \quad (t = k-1, \dots, 1).$$

Then

$$v_1 \longrightarrow v_2 \longrightarrow \dots \longrightarrow v_k$$

is a directed path containing exactly one vertex from each class. Consequently,

$$T(\pi, z) = \{v_1, \dots, v_k\}$$

is a transversal of $V_1, \dots, V_k$.

Consider the map

$$F: \mathcal{D} \longrightarrow \mathcal{T}, \quad F(\pi, z) = T(\pi, z),$$

where

$$\mathcal{D} = \{(\pi, z) : \pi \in \mathfrak{S}_k, z \in V_{\pi_k}\}$$

and $\mathcal{T}$ denotes the set of all transversals of the vertex classes. We claim that F is injective.

Suppose that

$$T(\pi, z) = T(\sigma, w) = T.$$

If $\pi \neq \sigma$, then the two distinct total orders induced on the classes disagree on the relative order of some pair of classes. Let $u, v \in T$ be the unique vertices belonging to those two classes. Interchanging u and v if necessary, we may assume that u occurs before v in the path associated with (π, z) , while v occurs before u in the path associated with (σ, w) .

The first path therefore contains a nonempty directed subpath from u to v , and the second contains a nonempty directed subpath from v to u . Concatenating them yields a nonempty closed directed walk contained entirely in T ; this remains true even if the two subpaths overlap. A shortest nonempty closed subwalk is a simple directed cycle. Since T contains at most one vertex from each class, this cycle is rainbow, contradicting the hypothesis.

It follows that $\pi = \sigma$. The two paths then have the same terminal class, and since T contains exactly one vertex from that class, their terminal vertices agree; hence $z = w$. Thus F is injective.

For each fixed permutation π , there are n_{π_k} possible terminal vertices. Moreover, each class occurs last in exactly $(k-1)!$ permutations. Hence

$$\begin{aligned} |\mathcal{D}| &= \sum_{\pi \in \mathfrak{S}_k} n_{\pi_k} \\ &= (k-1)! \sum_{i=1}^k n_i. \end{aligned}$$

On the other hand, the number of transversals is

$$|\mathcal{T}| = \prod_{i=1}^k n_i.$$

Injectivity of F now gives (1). □

Corollary 1. *Under the hypotheses of Theorem 1, if $n_i \leq d$ for every i , then*

$$k! \leq d^{k-1}. \tag{2}$$

Proof. Since all classes are nonempty, we may divide (1) by $\prod_{i=1}^k n_i$, which gives

$$\begin{aligned} 1 &\geq (k-1)! \sum_{i=1}^k \frac{n_i}{\prod_{j=1}^k n_j} \\ &= (k-1)! \sum_{i=1}^k \frac{1}{\prod_{j \neq i} n_j}. \end{aligned}$$

For every i ,

$$\prod_{j \neq i} n_j \leq d^{k-1},$$

so

$$1 \geq (k-1)! \frac{k}{d^{k-1}} = \frac{k!}{d^{k-1}}.$$

Rearranging proves (2). □

Corollary 2 (Linear upper bound). *For every positive integer d ,*

$$R(d) < ed.$$

Proof. Suppose that a graph witnessing $R(d) \geq k$ exists. By Corollary 1,

$$k! \leq d^{k-1}.$$

The standard strict factorial estimate

$$k! > \left(\frac{k}{e}\right)^k \tag{3}$$

holds for every positive integer k . Indeed, for $k \geq 2$, monotonicity of the logarithm gives

$$\log(k!) > \int_1^k \log x \, dx = k \log k - k + 1 > k \log k - k,$$

while the case $k = 1$ is immediate.

If $k \geq ed$, then (3) yields

$$k! > \left(\frac{k}{e}\right)^k \geq d^k \geq d^{k-1},$$

contradicting (2). Therefore every admissible k satisfies $k < ed$, and hence $R(d) < ed$. $\square$

5 The matching linear lower bound

For completeness, we include the standard construction of Chaudhury et al. [2].

Proposition 1. *For every positive integer d ,*

$$R(d) \geq d.$$

Proof. Take d vertex classes indexed by $i \in \{1, \dots, d\}$, with

$$V_i = \{(i, a) : a \in \mathbb{Z}/d\mathbb{Z}\}.$$

Thus every class has size exactly d .

For every pair $i \neq j$ and every $a \in \mathbb{Z}/d\mathbb{Z}$, include exactly the following edges:

$$(i, a) \longrightarrow \begin{cases} (j, a), & \text{if } i < j, \\ (j, a + 1), & \text{if } i > j, \end{cases} \quad (4)$$

where addition in the second coordinate is taken modulo d . There are no other edges.

We first verify the cross-part in-neighbor condition. Fix a target vertex $(j, b) \in V_j$ and a class V_i with $i \neq j$. If $i < j$, then $(i, b) \rightarrow (j, b)$ is an edge; if $i > j$, then $(i, b - 1) \rightarrow (j, b)$ is an edge. Thus every target vertex has an in-neighbor in every other class.

Suppose, for a contradiction, that this graph contains a rainbow cycle

$$(i_1, a_1) \longrightarrow (i_2, a_2) \longrightarrow \cdots \longrightarrow (i_r, a_r) \longrightarrow (i_1, a_1).$$

Set $i_{r+1} = i_1$ and $a_{r+1} = a_1$. Since the cycle is rainbow, the indices $i_1, \dots, i_r$ are distinct. By (4),

$$a_{t+1} - a_t \equiv \begin{cases} 0, & \text{if } i_t < i_{t+1}, \\ 1, & \text{if } i_t > i_{t+1} \end{cases} \pmod{d}.$$

Let

$$D = |\{t \in \{1, \dots, r\} : i_t > i_{t+1}\}|$$

denote the number of cyclic descents. Summing the label changes around the cycle gives

$$0 \equiv a_{r+1} - a_1 \equiv D \pmod{d}. \quad (5)$$

A cyclic sequence of $r \geq 2$ distinct indices has at least one ascent and at least one descent: the smallest index is followed by a larger one, while the largest index is followed by a smaller one. Consequently,

$$1 \leq D \leq r - 1 \leq d - 1,$$

which contradicts (5). Hence the graph contains no rainbow cycle.

When $d = 1$, the same construction consists of a single singleton class and no edges; the cross-part in-neighbor condition is vacuous, and there is no directed cycle. Thus the construction is valid for every positive integer d , proving $R(d) \geq d$. $\square$

Combining Corollary 2 and Proposition 1 gives the main conclusion.

Theorem 2 (Linear growth of the rainbow cycle number). *For every positive integer d ,*

$$d \leq R(d) < ed.$$

In particular,

$$R(d) = \Theta(d).$$

6 Scope of the result

The theorem resolves the conjectured linear growth of the rainbow cycle number, but it does not determine $R(d)$ exactly. In particular, the stronger conjecture

$$R(d) = d$$

remains open.

The counting argument is also fundamentally existential: by itself, it does not provide a deterministic polynomial-time procedure for finding a rainbow cycle at the linear threshold. Sivashankar [3] gives a separate randomized recovery argument. Finally, although the rainbow cycle number was introduced through an application to approximate EFX allocations, the combinatorial result proved here does not establish the existence of complete EFX allocations.

References

- [1] Bhaskar Ray Chaudhury, Jugal Garg, Kurt Mehlhorn, Ruta Mehta, Pranabendu Misra. *Improving Envy Freeness up to Any Good Guarantees Through Rainbow Cycle Number*. Mathematics of Operations Research, 2023. <https://doi.org/10.1287/moor.2021.0252>
- [2] B. R. Chaudhury, J. Garg, K. Mehlhorn, R. Mehta, and P. Misra, Improving envy freeness up to any good guarantees through rainbow cycle number, *Mathematics of Operations Research* 49 (2024), no. 4, 2323–2340. [doi:10.1287/moor.2021.0252](https://doi.org/10.1287/moor.2021.0252).
- [3] V. Sivashankar, A linear bound on the rainbow cycle number and approximate EFX, arXiv:2607.27455v1 [cs.GT], 2026. [doi:10.48550/arXiv.2607.27455](https://doi.org/10.48550/arXiv.2607.27455).